

Justin Manoj

(845) 633-1106 | justinmmanoj@gmail.com | [linkedin.com/in/justinmmanoj](https://www.linkedin.com/in/justinmmanoj) | github.com/justinm5

EDUCATION

University of Massachusetts – Amherst

Amherst, MA

B.S., Computer Science and Mathematics – GPA: 3.8/4.0

Expected Graduation: May 2027

- **Relevant Coursework:** Algorithms, Data Structures, Operating Systems, Database Systems, Machine Learning, Computer Networks & Security, Web Programming, Scalable Web Systems, Discrete Mathematics
- **Awards/Honors:** Commonwealth Honors College, Dean's List, Chancellor's Award, NYS Academic Excellence, National Merit Scholar, AP Scholar with Distinction

TECHNICAL SKILLS

Languages: Go, Python, Java, C/C++, JavaScript, TypeScript, SQL, HTML/CSS

Technologies: Spring Boot, FastAPI, React, Node.js, gRPC, PostgreSQL, Redis, Kafka, Docker, Kubernetes

Tools/Frameworks: AWS (EC2, EKS, S3), Linux/Unix, Git, GitHub Actions, CI/CD, OpenTelemetry

EXPERIENCE

IBM | Incoming Software Developer Co-op

Fall 2026

Dell Technologies | Software Engineering Intern

May 2026 – Present

- Designed and implemented an internal **Kubernetes** operator in **Go** using CRDs, admission webhooks, and reconciliation loops to automate audit logging across production clusters.
- Integrated **Syslog** and **OpenTelemetry Collector** pipelines to stream compliance events into a distributed observability platform, improving real-time security monitoring.
- Shipped **20+** production Jira stories including platform features, critical bug fixes, and CVE remediation across GitOps-managed **RKE2** clusters.

GBCS Group | Software Engineering Intern

March 2025 – September 2025

- Migrated **8** REST endpoints to a **GraphQL** API with DataLoader batching and **gRPC**-compatible internal services, cutting database queries by **60%** and reducing response payload sizes by **40%**.
- Reduced load times by **65%** using PostgreSQL composite indexes & Redis caching on high-traffic pages.
- Built **GitHub Actions** CI/CD pipeline with Jest test suites and Sentry monitoring, reducing production bugs reported per sprint by **35%**.

University of Massachusetts Amherst | Undergraduate Research Assistant

November 2024 – April 2025

- Stress-tested **7** multi-agent LLM frameworks uncovering **14** failure patterns across **10,000+** simulation runs.
- Sped up log processing **3×** with Ray & Pandas, analyzing **1,600+** execution traces to benchmark agent reliability.
- Cut review time by **70%** with an automated pipeline that classifies agent failures and surfaces critical patterns.

BUILD UMass | Software Engineer

February 2024 – May 2024

- Indexed **1,500+** campus events from **300+** student organizations using **Elasticsearch**, reducing search latency by **89%** and supporting real-time filtering by date, category, and location.
- Engineered real-time event updates with **WebSockets** and **Redis Pub/Sub**, delivering live notifications to concurrent users with **sub-50ms** delivery time.
- Worked with a team of **4** to ship to **6,000+** active UMass students, processing **500+** daily searches in the semester.

PROJECTS

Quantitative Research Agent | Go, gRPC, TimescaleDB, Kafka, Redis, Python

Jun. 2026

- Designed an algorithmic-research platform ingesting SEC Form 4 filings, live price feeds, and financial news streams into a **Go/gRPC/Kafka** microservices pipeline with **TimescaleDB** and **Redis**.
- Trained **Python** anomaly-detection agent to flag filing/price divergences and serve signals through a React UI.

UMass Dining Engine | Java, Spring Boot, React, PostgreSQL, Redis, Python

Jan. 2026

- Shipped an AI meal planning platform for **30,000+** UMass students aggregating live menus from **4** dining halls with real-time dietary filtering using **Redis-backed Spring Boot API** and **PostgreSQL** text-search.
- Built a React frontend and scheduled **Python ETL**; pending adoption into official UMass Dining App.